Timeline Check-in

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Set up

Packages

```r
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# arranging plots
library(patchwork)
library(ggridges)
```

Data

```r
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")
```

```r
# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
↪  Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning and Wrangling

Cleaning

`sites` object

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

`taxon_list` object

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake")) |>
  select(verbatim_name, scientificName)
```

`water_quality` object

```r
water_quality_clean <- water_quality |>
  clean_names() |>
  semi_join(ncos_sites, by = "site") |>
  unite("date_site", date, site, remove = TRUE) |>
  group_by(date_site) |>
  summarize(med_ph    = median(p_h, na.rm = TRUE),
```

```
            med_sal   = median(salinity_ppt, na.rm = TRUE),
            med_do    = median(dissolved_oxygen_mg_l, na.rm = TRUE),
            med_temp  = median(temperature_c, na.rm = TRUE)) |>   # <-- ADD
             ↪  THIS
  ungroup() |>
  filter(date_site != "2022-08-12_NVBR")
```

`aq_ins` object

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         cladocera) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:cladocera,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5
   ↪  liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
```

```
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
   ↪  = TRUE),
         site_full = fct_rev(site_full)) |>

  # ADD: log+1 transform of average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>                        # <-- Fig
   ↪  1, 2, 3, 4

  # ADD: season derived from date
  mutate(season = case_when(
    month(date) %in% c(3, 4, 5)  ~ "Spring",
    month(date) %in% c(6, 7, 8)  ~ "Summer",
    month(date) %in% c(9, 10, 11) ~ "Fall",
    month(date) %in% c(12, 1, 2) ~ "Winter"
  ),
  season = factor(season, levels = c("Spring", "Summer", "Fall", "Winter")))
   ↪  |># <-- Fig 4
  mutate(taxon = factor(taxon, levels = c("copepod", "ostracod",
   ↪  "cladocera")))

#ADD: capitalizing invertebrate names
aq_ins_clean$taxon <- str_to_sentence(aq_ins_clean$taxon)
```

Setting plot aesthetics

```
# colors for taxa
taxon_cols <- c(
  "ostracod" = "orange2",
```

```
  "copepod" = "tomato3",
  "cladocera" = "turquoise4")

sites_cols <- c(
  "NDC" = "hotpink3",
  "NEC" = "cyan3",
  "NMC" = "lightpink2",
  "NPB" = "slateblue3",
  "NPB1" = "steelblue3",
  "NPB2" = "green3",
  "NVBR" = "maroon3"

)

# setting common theme elements
theme_set(
  theme_classic() +
    theme(text = element_text(size = 12),
          plot.title.position = "plot",
          legend.position = "none"))
```

Visualizations directly relevant to answering questions

Figure 1

```
# create new object to store figure 1
figure1 <- ggplot(data = aq_ins_clean,
                  mapping = aes(x = med_temp,
                                y = log_ave_lit,
                                color = taxon)) +
  geom_point() +

  facet_wrap(~site_full) +
  labs(title = "Figure 1:

Dominant Three Taxa Abundance and Average Water Temperature by Site",
       x = "Average Temperature (ºC)",
       y = "log + 1 Transformed Average Abundance/L",
       color = "Taxon") +
```

```
  scale_color_manual(values = taxon_cols) +

  theme(legend.position = "right")



figure1
```

Figure 1:

Dominant Three Taxa Abundance and Average Water Temperature by Site

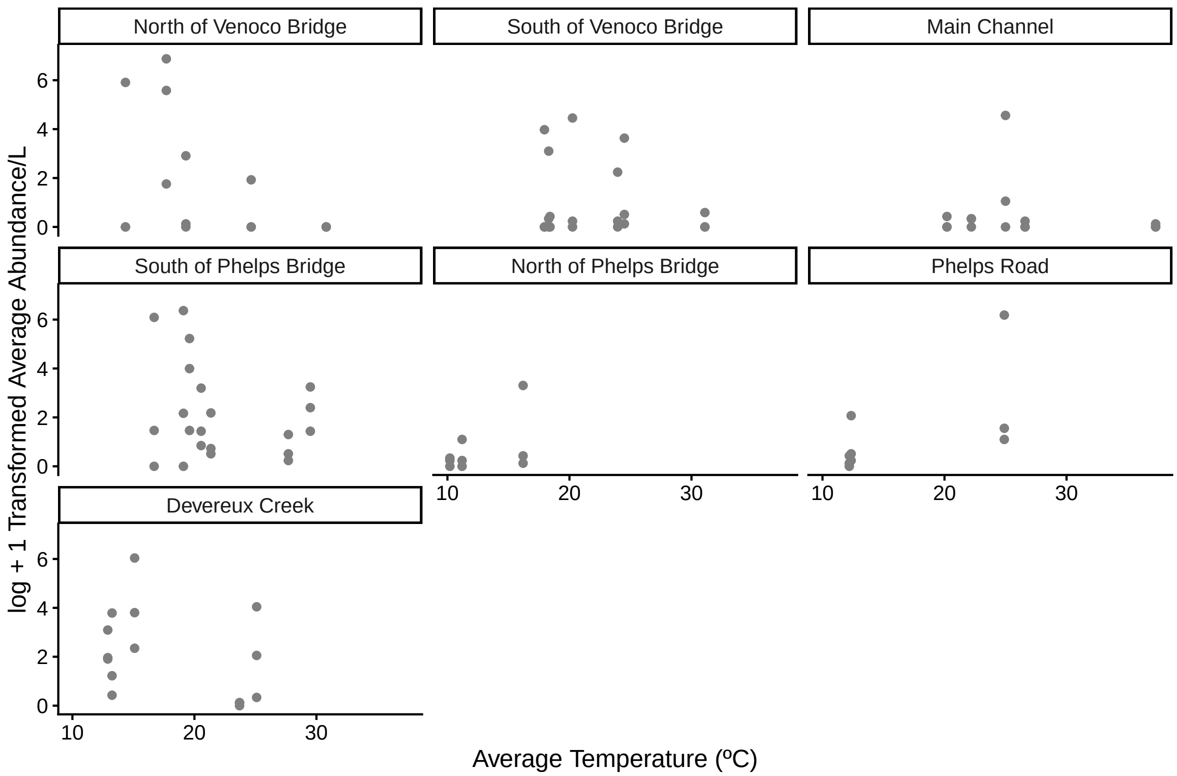


Figure 2

```
# create new object to store figure 2
figure2 <- ggplot(data = aq_ins_clean,
                  aes(x = taxon,
                      y = log_ave_lit,
```

```
                     color = taxon)) +

  geom_jitter(height = 0,
              alpha = 1,
              width = 0.2,
              shape = 21) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "mean",
               size = 4,
               color = "black",
               shape = 18) + # 18 = filled diamond

   stat_summary(geom = "errorbar",
               fun.data = mean_se,
               width = 0.1,
               color = "black") +

  scale_color_manual(values = taxon_cols) +
  labs(x = "Taxon",
       y = "log + 1 transformed average abundance/L",
       title = "Figure 2:

Mean and Distribution of Average Abundance for Three Dominant Taxa")

figure2
```

Figure 2:

Mean and Distribution of Average Abundance for Three Dominant Taxa

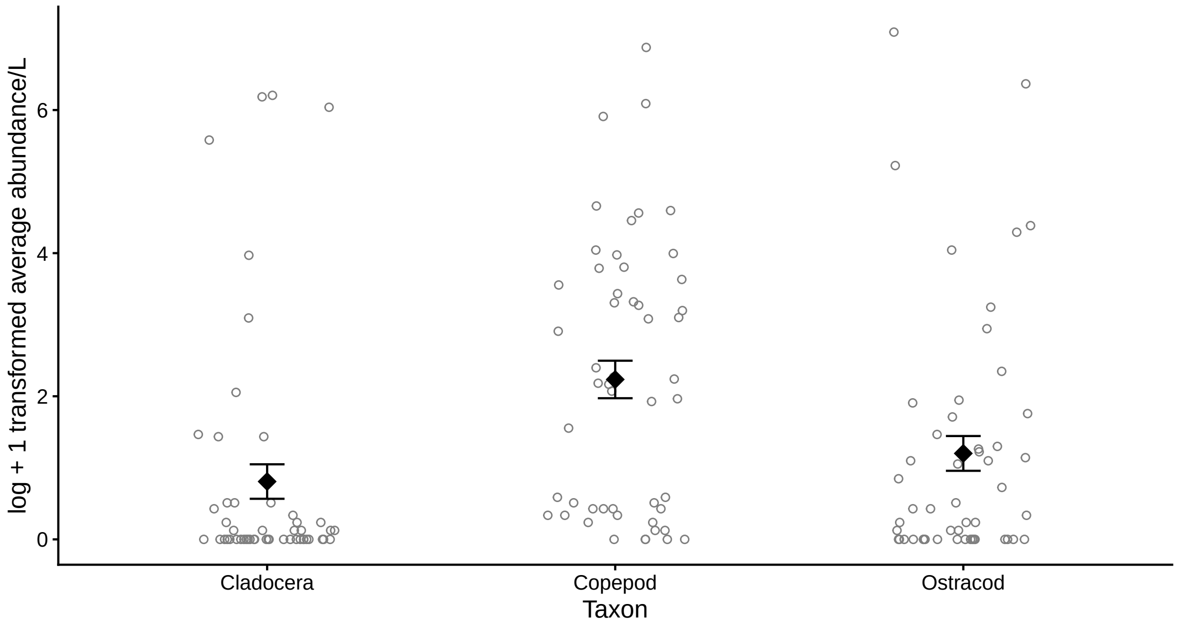


Figure 3a

```
# create new object to store figure 3a
figure3a <- ggplot(data = aq_ins_clean,
                   mapping = aes(x = med_temp,
                                 y = log_ave_lit)) +

  geom_point() +

  facet_wrap(~taxon,
             scales = "free") +

  theme(legend.position = "right")+

  labs(title = "Figure 3a:

Average Abundance and Average Water Temperature for Three Dominant Taxa",
       x = "Average Temperature (ºC)",
       y = "log + 1 Transformed Average Abundance/L",
```

```
       color = "Site")

figure3a
```

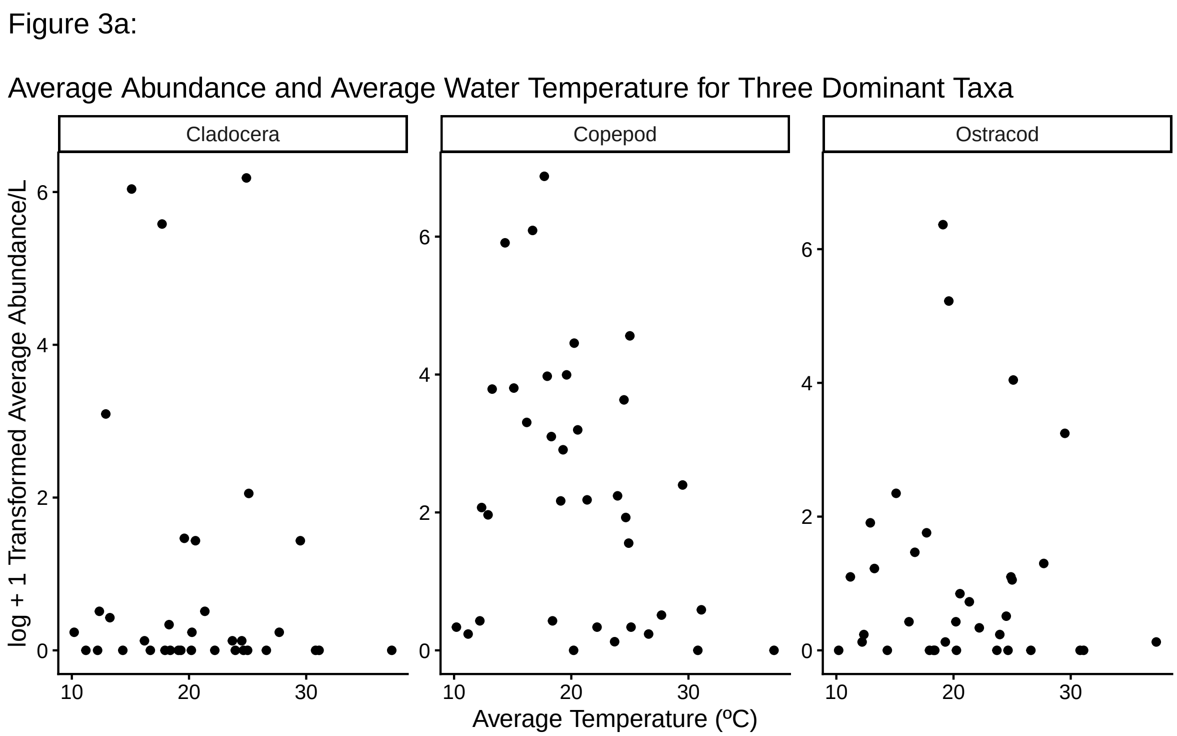


Figure 3b

```
# create new object to store figure 3b
figure3b <- ggplot(data = aq_ins_clean,
                   mapping = aes(x = site,
                                 y = log_ave_lit,
                                 color = site)) +

  geom_point() +

  facet_wrap(~taxon) +

  theme(legend.position = "right") +
```

```
  scale_color_manual(values = sites_cols) +
  labs(title = "Figure 3b:
Average Abundance and Average Water Temperature for Three Dominant Taxa by
↪  Site",
       x = "Average Temperature (ºC)",
       y = "log + 1 Transformed Average Abundance/L",
       color = "Site")

figure3b
```

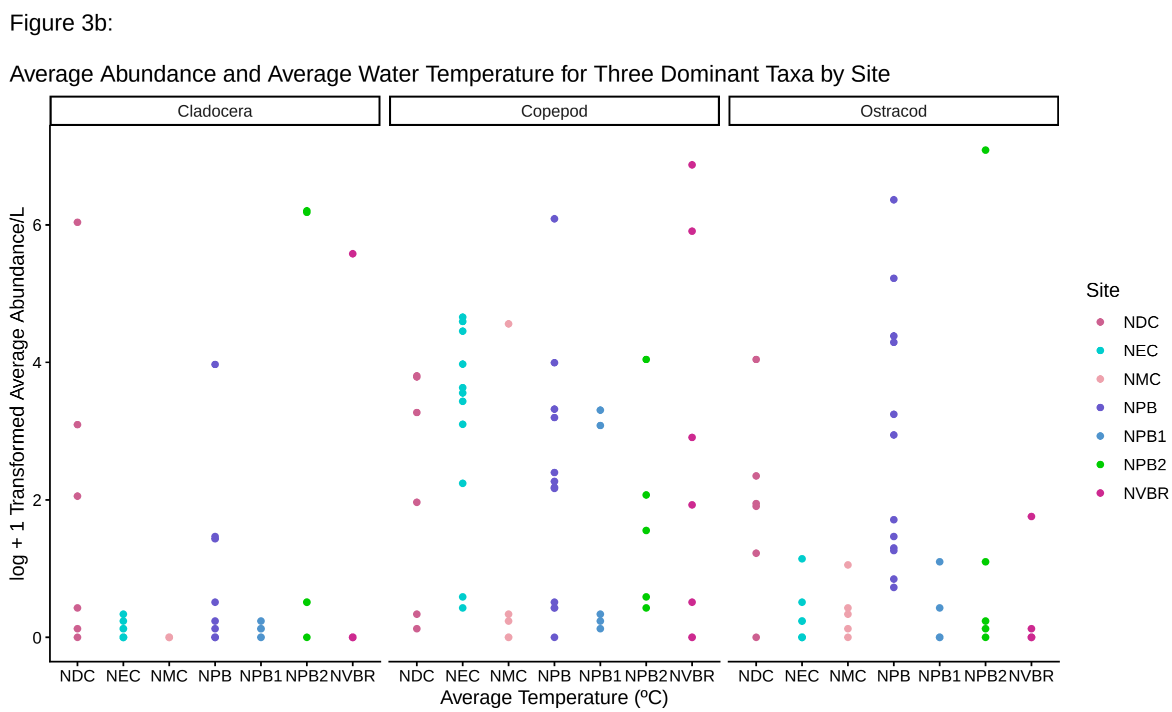


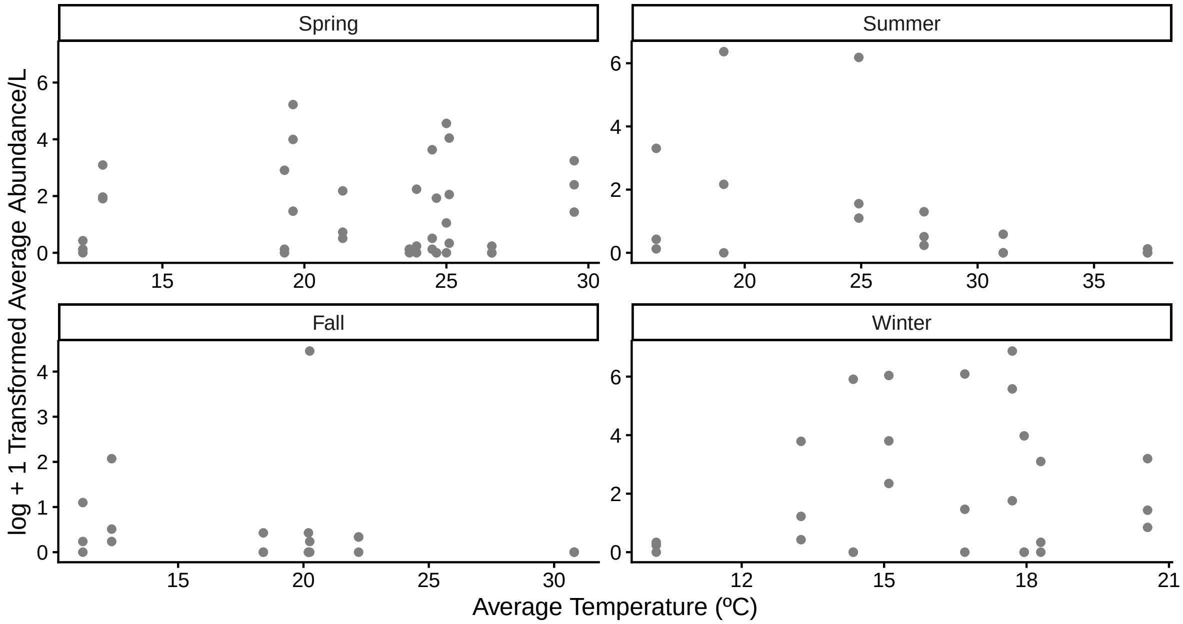
Figure 4

```
# create new object to store figure 4
figure4 <- ggplot(data = aq_ins_clean,
                  mapping = aes(x = med_temp,
```

```
                                 y = log_ave_lit,
                                 color = taxon)) +

  geom_point() +

  facet_wrap(~season,
             scales = "free") +
  labs(title = "Figure 4:

Average Abundance and Average Water Temperature by Season for Three Dominant
 ↪  Taxa",
       x = "Average Temperature (ºC)",
       y = "log + 1 Transformed Average Abundance/L",
       color = "Taxon") +

  scale_color_manual(values = taxon_cols) +

  theme(legend.position = "right")


figure4
```

Figure 4:

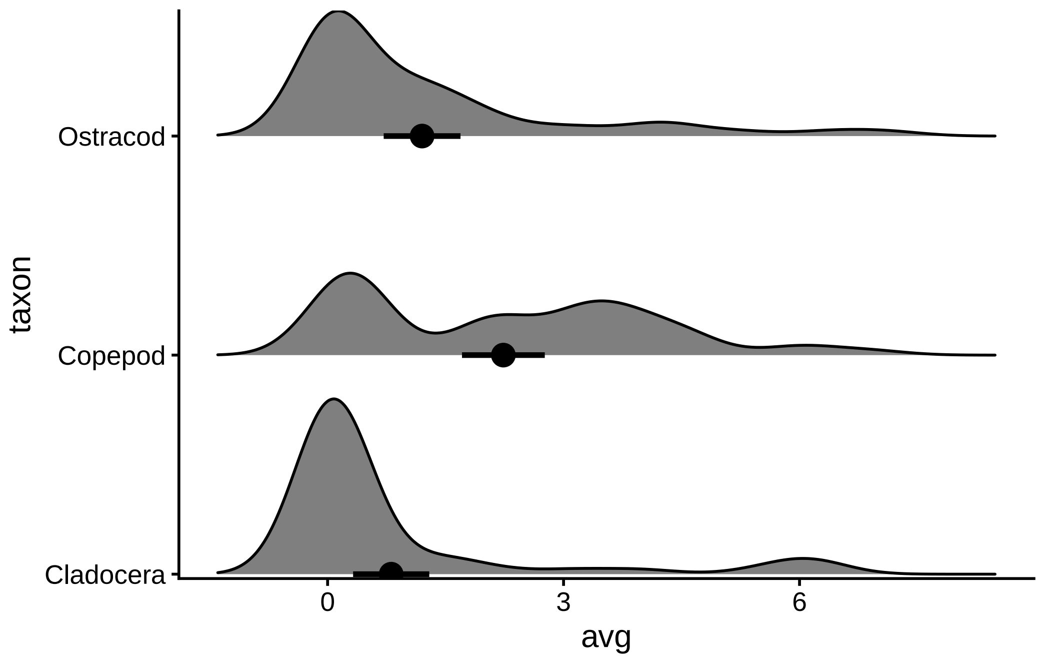
Average Abundance and Average Water Temperature by Season for Three Dominant Taxa



Other exploratory visualizations

```
# creating new object for ridgeline
ridges <- ggplot(data = aq_ins_clean,
                 # x-axis: avg temp
                 # y-axis: avg abund
                 # fill geometries by site
                 mapping = aes(x = log_ave_lit,
                               y = taxon,
                               fill = taxon)) +
  # ridges (scale makes ridges shorter so that they don't overlap)
  geom_density_ridges(scale = 0.8) +

  # adding means and 95% confidence intervals
  stat_summary(geom = "pointrange",
               fun.data = mean_cl_normal,
               size = 0.75,
               linewidth = 1) +
  # filling geometries with color
```

```
  scale_fill_manual(values = sites_cols) +
  # setting y-axis expansions
  scale_y_discrete(expand = expansion(mult = c(0.01, 0.15))) +
  # relabelling x- and y-axis
  labs(x = "avg",
       y = "taxon")

# displaying object
ridges
```



Ideas of Analysis for Background

-

Plan for elective

Timeline

Week 7:

- Research all three organisms (copepod, ostracod, cladoceran)

- Gather reference images
- Review your figures
- Take notes on what each figure shows about the species
- Brainstorm collage/drawing style and decide on materials (digital vs. physical).

Week 8 (timeline check-in):

- Have a rough sketch/draft of all three 2D collages– one per organism
- Bring your notes on what the figures show about each species. This is our plan and draft milestone.

Week 9:

- Refine and finalize the artwork for 1–2 of the organisms
- Fill in your species notes with complete observations from your figures.

Week 10:

- Complete the third organism's artwork and finish all species notes
- Do a full review, make sure each collage clearly reflects what your figures show about that species.

Finals week:

- Final polish, compile everything together
- Present your completed 2D collage triptych with notes.

Visualization Approach and Temperature Color Scale

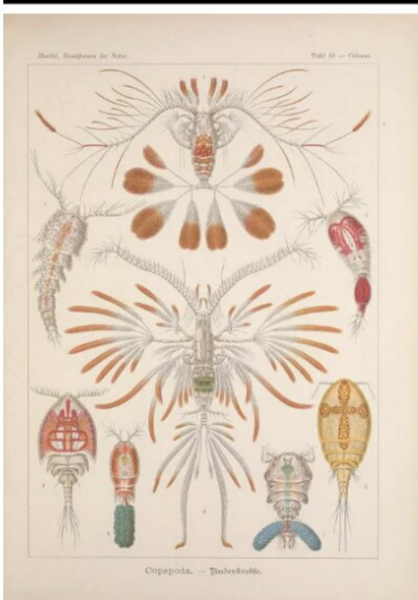
For our final visualization, we decided on a physical drawing. Each organism will be represented by an outline, filled with paint and pastels to illustrate the relationship between abundance and temperature. The colors inside each outline correspond to a temperature-based heat range, divided into five equal intervals spanning the full range of recorded temperatures (10.20°C – 37.30°C):

- 10.0°C – 15.6°C
- 15.6°C – 21.2°C
- 21.2°C – 26.8°C
- 26.8°C – 32.4°C
- 32.4°C – 38.0°C

Gathering Images and Research on Organisms

Copepods

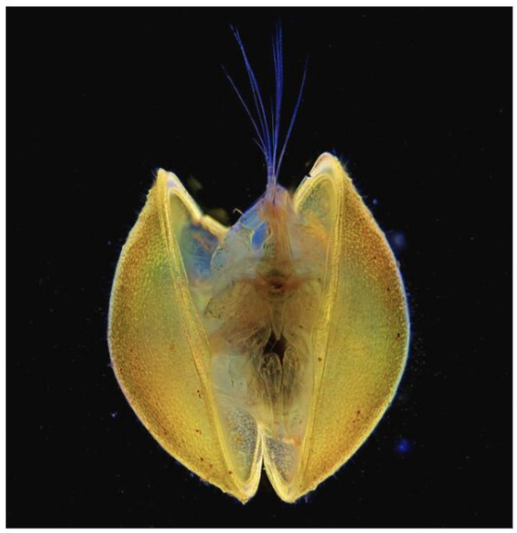
- Group of small crustaceans
- Found in nearly every salt and freshwater habitat
- Some are planktonic, some benthic (live on sediments)
- Teardrop shaped body with large antennae
- Thin exoskeleton is transparent; they eat phytoplankton, detritus, and small zooplankton
- Several species are bioluminescent
- They jump out of the water to evade predators



Ostracods

- tiny, ubiquitous crustaceans ranging from 0.1 to 30 mm in size
- Commonly known as “seed shrimp”
- Bivalve shells that encase their bodies
- About 33,000 species, some being bioluminescent
- Most common arthropods in the fossil record



Cladocera

- Commonly known as the water flea
- Freshwater crustaceans, feeding on microscopic organics, though some forms are predatory
- Over 1000 species
- Down-turned head with a single median compound eye, and a carapace covering the apparently unsegmented thorax and abdomen
- Most cladoceran species live in fresh water and other inland water bodies